Data Analysis Plan:
Linear Regression Model for Concrete Compressive Strength

1. Introduction

As the cornerstone of the construction industry, concrete is widely used in the core structures of various infrastructure projects such as houses, bridges and roads. By analyzing and enhancing its compressive strength, not only can the structural safety be ensured, but also the material usage can be optimized to achieve higher economic efficiency and durability.

In this report, we outline our intended methodology and steps for applying a linear regression model to (1) explain how each ingredient and curing age influences concrete strength and quantify these relationships and (2) predict the compressive strength of a new mix based on its ingredient proportions and age.

Our goal is to provide clear, actionable recommendations based on data-driven insights that can help inform decisions regarding mix design optimization for ready-mix concrete producers seeking to reliably meet required strength targets while controlling costs and minimizing material waste.

2. Data Description

The compressive strength depends on how much cement, blast furnace slag , fly ash, water, superplasticizer, coarse aggregate, fine aggregate and the curing age are used in the mixture. The Concrete Compressive Strength data set from Prof. I-Cheng Yeh contains measurements of these ingredients together with the resulting compressive strength for 1030 different mixtures. After conducting background research on each component, the specific information is as follows in the table.

Component	Role	"Common Sense" Rules
Cement	The core cementitious material, which reacts with water to form a gel that binds the aggregates.	Positive
Blast Furnace Slag	An industrial by-product that improves long-term strength and durability, and reduces heat of hydration. It can partially replace cement and contributes significantly to the later strength.	Nonlinear
Fly ash	An industrial by-product that improves workability and participates in the reaction later to form strength. It can reduce costs, but has lower early strength.	Nonlinear
Water	Participates in the hydration reaction, giving construction workability.	Negative
Superplasticizer	Reduces water usage while maintaining fluidity, or increases slump while not adding water. An increase in dosage can lower the water-cement ratio, thereby increasing strength.	Positive

Coarse aggregate	The concrete framework, bearing the main load.	Positive
Fine aggregate	Fills the voids, improving density	Positive
Age	The 28-day strength is the benchmark for project acceptance.	Positive

Apart from the intensity, the selection of each component is also driven by multiple factors. (1)Economy. The cost of cement is the highest. Enterprises tend to minimize its usage. While slag and fly ash, as cheap industrial by-products, are preferred to be maximally incorporated to replace cement and reduce costs. (2)Construction. To ensure smooth pumping at the site, workers sometimes increase the water addition to the maximum extent on their own; the water reducer improves the fluidity by maximizing the dosage without increasing the water content.
(3)Durability. The maximum addition of slag and fly ash can enhance the resistance to erosion and prolong the service life of the structure; minimizing the water-cement ratio significantly reduces permeability and improves durability.

3. Methodology

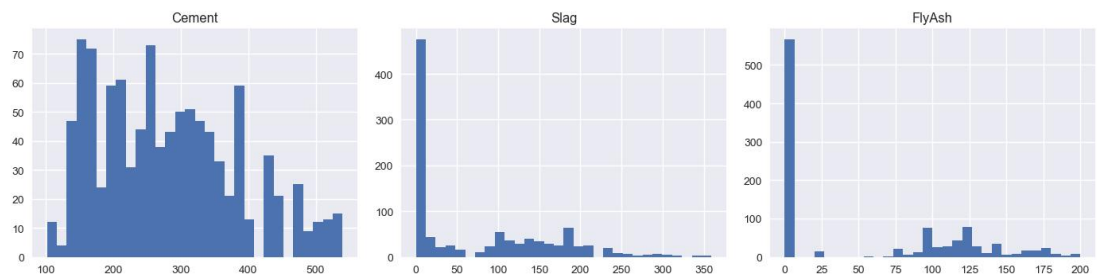
3.1 Data Exploration

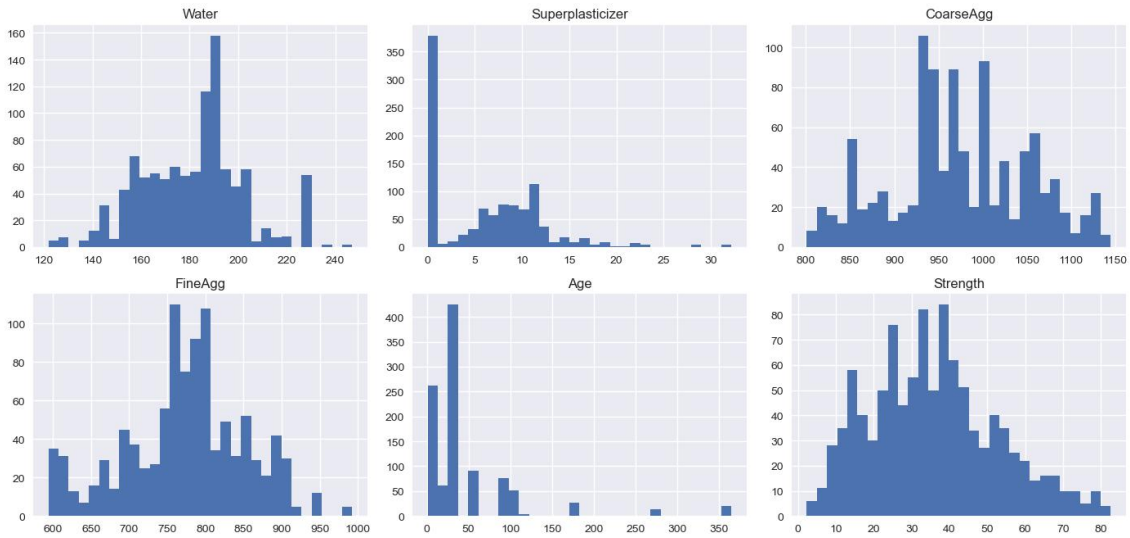
An initial exploratory data analysis has been performed to understand the distribution and relationships between variables.

(1)Summary Statistics

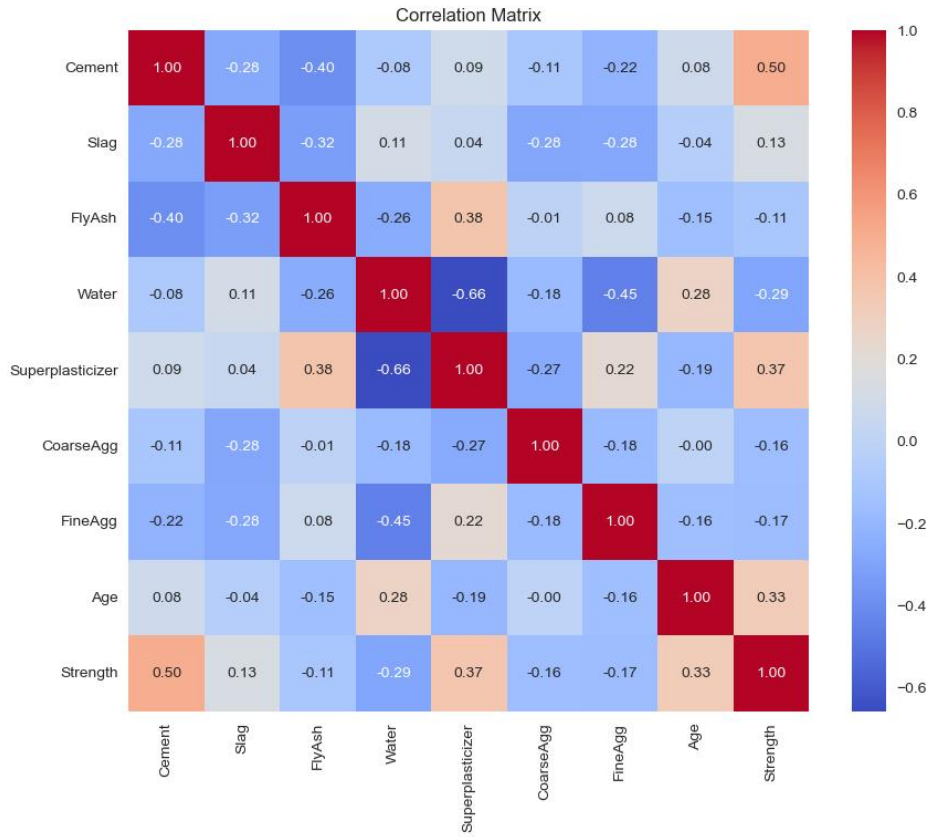
	mean	std	min	max	range
Cement	281.17	104.51	102.00	540.00	438.00
Slag	73.90	86.28	0.00	359.40	359.40
FlyAsh	54.19	64.00	0.00	200.10	200.10
Water	181.57	21.35	121.80	247.00	125.20
Superplasticizer	6.20	5.97	0.00	32.20	32.20
CoarseAgg	972.92	77.75	801.00	1145.00	344.00
FineAgg	773.58	80.18	594.00	992.60	398.60
Age	45.66	63.17	1.00	365.00	364.00
Strength	35.82	16.71	2.33	82.60	80.27

(2)Distributional Analysis





(3)Correlation Matrix



According to the histogram, multiple variables do not follow a strictly normal distribution. Among them, Slag, Fly Ash, Superplasticizer, and Age show a significant right skew, and some variables are highly concentrated around 0, indicating that they were not used in some combinations. Table 1 shows that the scale differences of each variable are significant, especially the fluctuation range of Age is much larger than that of other variables, which indicates the necessity of standardization processing.

The correlation matrix indicates that Cement and Strength have the strongest positive correlation (r = 0.50), Water and Strength have a negative correlation (r = -0.29), and Superplasticizer and Age also have a moderate positive correlation. Additionally, there is a strong negative correlation between Water and Superplasticizer (r = -0.66), suggesting a certain degree of structural correlation, but it has not yet reached the level of severe multicollinearity.

3.2 Model Building

A linear regression model will be fit to the training data:

Concrete_Strength = m_1 * Cement + m_2 * Blast_Furnace_Slag + m_3 * Fly_Ash + m_4 * Water + m_5 * Superplasticizer + m_6 * Coarse_Aggregate + m_7 * Fine_Aggregate + m_8 * Age + b

This model will be evaluated using the R-squared performance metric on both the training and testing datasets.

Optimization of model parameter values will be conducted using both Ordinary Least Squares, as well as our own implementation of gradient descent.

4. Expected Deliverables

• A comprehensive report detailing the methodology, results, and conclusions.
• The final linear regression model and an analysis of its reliability.
• Recommendations for using the model in practical applications.

Based on the current analysis results, for example, if the regression model further confirms that Cement has a significant positive impact on strength and Water has a negative correlation, it is recommended to optimize the water-cement ratio while meeting the workability requirements, and reduce the water usage by reasonably using Superplasticizer, thereby enhancing the strength and controlling the cost. For high-strength demand structures, the proportion of cement can be moderately increased; for general engineering projects, part of the cement can be replaced by Slag or Fly Ash, while meeting the strength standards, to reduce material costs and carbon emissions.

Furthermore, it is recommended to integrate the regression model into the enterprise matching design process. Before the experiment, perform intensity prediction to achieve pre-assessment of matching and quality control. This will help reduce the cost of experiments, improve production efficiency, and enhance the company's ability to balance safety and economy.